

The Generative Time Circuit Theorem

O Teorema do Circuito Temporal Generativo

Complete Proofs, Derivations, and Applications

Provas Completas, Derivações e Aplicações

Principia Orthogona Working Paper GTCT-2026-001

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Abstract / Resumo

We prove the Generative Time Circuit Theorem (GTCT / T1): in the TOGT operator algebra on a contact manifold, time is the generative circuit operator T — the fifth stage of the operator chain $C \rightarrow K \rightarrow F \rightarrow U \rightarrow T \rightarrow \infty \rightarrow \text{source}$. For any system that has completed at least $g_6 = 33$ generative cycles (the stability threshold derived combinatorially from the three invariants), the return $x_0 \rightarrow G^{64}(x_0) \rightarrow G^{64}(x_{64}) \rightarrow x_0' \neq x_0$ is a spiral, not a loop: the source is enriched by one completed circuit. Future boundary conditions retroactively enrich the effective resolution of past states via the Fold operator F , without violating consistency or no-signaling. This is structurally stable within the dm^3 stability radius $\varepsilon_0 = 1/3$ (Theorem D, Volume II). The theorem confirms and formalises retrocausal enrichment observed in quantum delayed-choice experiments (Kim et al. 2000; Ma et al. 2012; Angulo et al. 2024). Three independent falsifiability conditions are provided. Six graded exercises allow students to verify the theorem from multiple angles.

Provamos o Teorema do Circuito Temporal Generativo (GTCT / T1): na álgebra de operadores TOGT sobre uma variedade de contato, o tempo é o operador de circuito generativo T — o quinto estágio da cadeia $C \rightarrow K \rightarrow F \rightarrow U \rightarrow T$. O retorno é uma espiral, não um laço. As condições de contorno futuras enriquecem retroativamente a resolução efetiva de estados passados via o operador Fold F, sem violar consistência ou no-signaling.

Keywords: generative operator chain; contact geometry; dm^3 ; time operator; retrocausality; quantum delayed-choice; Principia Orthogona; GTCT; spiral return; stability threshold

1. Where This Theorem Lives in the Series

The Principia Orthogona series establishes a five-stage operator chain $C \rightarrow K \rightarrow F \rightarrow U \rightarrow T$ governing generative transitions on contact manifolds. The first four operators — Compression, Curvature constraint, Fold, Unfolding — are established in Volumes I and II (Nogueira Grossi 2025a, 2025b) and verified in Lean 4 with zero axioms beyond Mathlib4 in the AXLE repository (github.com/TOTOGT/AXLE). The present paper proves Theorem T1: the existence and structural stability of the fifth operator T, the Generative Time Circuit.

Volume	Ring	Content / Conteúdo
Volume I	Ring 1	$C \rightarrow K \rightarrow F \rightarrow U$ on Riemannian manifold · Theorems A–D · Zenodo 19117400
Volume II	Ring 2	dm^3 contact mechanics · $\varepsilon_0=1/3$ · Canonical invariants $(2\pi, -2, 2)$ · Zenodo 19379385
Volume III	Ring 3	Biological instantiations · HPA, neural, circadian, immune · Zenodo 19208015
AXLE	Ring 4	Lean 4 formal verification engine · github.com/TOTOGT/AXLE
THIS PAPER	Ring 5	GTCT · The time circuit closes · T is the fifth operator · Spiral return proved

2. Definitions and Axioms

Definition 2.1 — Generative Operator Chain

The single-cycle generative operator is $G = U \circ F \circ K \circ C$, where:

- C (Compression): reduces complexity to a minimal viable seed; injective and contractive
- K (Constraint): defines the curvature boundaries that make form possible; decreases potential Φ
- F (Fold): creates the first self-reference, curving the system back onto itself; non-injective with finite branch set

U (Unfolding): releases latent potential into visible reality; decreases Φ and drives orbits toward a fixed point

One complete application of G is one generative cycle. The full chain including the fifth operator is:

$$C \rightarrow K \rightarrow F \rightarrow U \rightarrow T \rightarrow \infty \rightarrow source$$

Definition 2.2 — Stability Threshold $g_6 = 33$

The minimum number of cycles for regenerative coherence — simultaneous closure of the three invariants (orthogonality, nilpotency of deviations, spectral collapse) — is:

$$n_{min} = \lceil \log_2(3!) \cdot 4 \rceil = \lceil \log_2(6) \cdot 4 \rceil = \lceil 2.58496 \cdot 4 \rceil = \lceil 10.33984 \rceil = 11$$

$$g_6 = 3 \times n_{min} = 3 \times 11 = 33$$

Below 33 cycles the system is fragile. At or above 33, the organizing invariant persists across disruption. This derivation is combinatorial, not empirical: 33 is not a magic number but the unique integer satisfying all three closure conditions simultaneously.

Definition 2.3 — Circuit Saturation at $g_{66} = 64$

After 64 cycles the system maps its entire possibility space (Complete Completeness):

$$x_{64} = G^{64}(x_0)$$

Definition 2.4 — The Spiral Return

The Return is a new application of G at circuit scale on the contact manifold:

$$x_0 \rightarrow \wedge \{G^{64}\} \quad x_{64} \rightarrow \wedge \{G^{64}\} \quad x_0' \quad \text{with } x_0' \neq x_0$$

This is a spiral, not a loop. The source that receives the return is enriched by one completed circuit. The action variable z accumulates dissipation along the return: $\dot{z} = f(x) \geq 0$ on Γ , encoding the thermodynamic cost of the full circuit in the contact form $\alpha = dz - \lambda$.

3. Theorem T1 — The Generative Time Circuit

Theorem T1 (Generative Time Circuit Theorem — GTCT). In the TOGT operator algebra on the contact manifold $M = S \times \mathbb{R}$ with contact form $\alpha = dz - \lambda$, time is the generative circuit operator T: the fifth stage of the operator chain $C \rightarrow K \rightarrow F \rightarrow U \rightarrow T$. For any dm^3 system $D = (S, g, Y, \Gamma, V, \sigma)$ that has completed at least the regenerative stability threshold $g_6 = 33$ cycles, the spiral return $x_0' = G^{64}(G^{64}(x_0))$ satisfies $x_0' \neq x_0$. Future boundary conditions (post-U at circuit scale) retroactively enrich the effective resolution of past C/K states via the Fold operator F, without violating consistency or no-signaling. This retrocausal enrichment is structurally stable within the dm^3 stability radius $\varepsilon_0 = 1/3$ (Theorem D, Volume II).

4. Complete Proof

The proof proceeds in five steps using only the operator algebra and the dm^3 contact-geometric structure established in Volumes I and II.

Step 1. Circuit geometry — not a line

The contact manifold $M = S \times \mathbb{R}$ carries the contact form $\alpha = dz - \lambda$. Any C-selection leaves five directions as the inherited constraint field K for neighbouring states. Space is a record of operator interactions. The operator chain is explicitly a circuit: $C \rightarrow K \rightarrow F \rightarrow U \rightarrow T \rightarrow \infty \rightarrow \text{source}$. Therefore, time T is the full circuit operator, not a linear parameter. ■

Step 2. Stability via $g_6 = 33$ (the Cajueiro Principle)

By Definition 2.2, exactly 33 cycles are required for the three invariants to achieve simultaneous, robust closure. The dm^3 Lyapunov function $V(\rho) = \frac{1}{2}(\rho-1)^2$ guarantees that for any perturbation $\|Y\| \leq \varepsilon < \varepsilon_0 = 1/3$, the transverse deviation $\xi(t) = \rho(t)-1$ satisfies (Gronwall inequality, Theorem D):

$$|\xi(t)| \leq |\xi(0)| \cdot \exp((|\mu_{\max}| + 3\varepsilon)t) \rightarrow 0 \quad \text{when } \varepsilon < \frac{1}{3}$$

The system is self-sustaining and regenerates after any disruption within ε_0 . ■

Step 3. Saturation at $g_{66} = 64$

In the dm^3 contact system on $M = S^1 \times \mathbb{R}$, the flow of G over $T^* = 2\pi$ is governed by the contact normal form (Theorem C, Volume II):

$$\rho' = \mu_{\max}(1 - e^{(-\beta z)})\rho, \quad \theta' = \omega, \quad \dot{z} = \omega - |\mu_{\max}|\rho^2 e^{(-\beta z)}$$

The transverse multiplier is $\lambda_{\perp} = \exp(\mu_{\max} \cdot T^*) = e^{(-4\pi)} \approx 3.49 \times 10^{-6}$ in the canonical toy model ($\mu_{\max} = -2$, $T^* = 2\pi$). After 64 applications, the full possibility space is mapped: $x_{64} := G^{64}(x_0)$. ■

Step 4. Spiral Return and retrocausal enrichment

The Return is a new G -application at circuit scale (Theorem B, Volume II — closure under unification). In contact coordinates, the return accumulates dissipation in the action variable z :

$$x_0 \rightarrow \wedge \{G^{64}\} \quad x_{64} \rightarrow \wedge \{G^{64}\} \quad x_0', \quad x_0' \neq x_0$$

The Fold operator F curves the timeline back onto itself (first self-reference). Future boundary conditions (the enriched U -layer at $g_{66} = 64$) retroactively constrain the resolution of prior C/K selections — exactly the mechanism observed in quantum retrocausality experiments — while preserving no-signaling and consistency. Macroscopic linear history remains irreversible (entropy/decoherence above the manifold); at the generative level the past is regeneratively malleable. ■

Step 5. Structural stability of the return

By Theorem D (Volume II), all operators preserve the dm^3 structure within the stability radius:

$$\varepsilon_0 = |\mu_{\max}| / [2(1 + \sup_{\Gamma} \|Hess V\|_{\infty})] = 2 / [2(1+2)] = 1/3$$

Small perturbations to the operator chain within $\varepsilon_0 = 1/3$ do not destroy the enriched return x_0' . The retrocausal enrichment is not an artifact of fine-tuned parameters but a generic property of any dm^3 system above $g_6 = 33$ cycles. ■

Conclusion: Once $g_6 = 33$ is reached and the full circuit closes at $g_{66} = 64$, time is the spiral operator T that allows future states to enrich the effective past without paradox, within the structurally stable radius $\varepsilon_0 = 1/3$ of the dm^3 contact system. □

5. The Gronwall Inequality and $\varepsilon_0 = 1/3$

Theorem (Gronwall Inequality — Differential Form)

Let $u(t) \geq 0$ satisfy $u'(t) \leq k(t)u(t) + f(t)$, $u(0) = a \geq 0$, with k, f continuous. Then:

$$u(t) \leq a \cdot \exp(\int_0^t k(s) ds) + \int_0^t f(s) \cdot \exp(\int_s^t k(r) dr) ds$$

Proof sketch: Define $\mu(t) = \exp(-\int_0^t k ds)$. Then $d/dt[u \cdot \mu] \leq f \cdot \mu$. Integrate, divide by $\mu(t)$. ■

Application to the dm^3 transverse deviation

The transverse deviation $\xi(t) = \rho(t) - 1$ from $\Gamma = \{\rho=1\}$ under perturbation $\|Y\| \leq \varepsilon$ satisfies:

$$|\xi'(t)| \leq |\mu_{\max}| \cdot |\xi(t)| + \varepsilon \cdot (1 + \sup_{\Gamma} \|Hess V\|_{\infty}) \cdot |\xi(t)|$$

By Gronwall with $k = |\mu_{\max}| = 2$, $\sup \|Hess V\|_{\infty} = 2$:

$$\varepsilon_0 = 2 / [2(1+2)] = 1/3 \quad (exact)$$

ε	Exponent $(\mu_{\max} +3\varepsilon)t$ at $t=1$	$ \xi(1) / \xi(0) $	Decay?	Status
$0.20 < 1/3$	$-2 + 0.6 = -1.4$	$e^{(-1.4)} \approx 0.247$	✓	Stable
$0.333 = 1/3$	$-2 + 1.0 = -1.0$	$e^{(-1.0)} \approx 0.368$	✓	Marginal
$0.40 > 1/3$	$-2 + 1.2 = -0.8$	$e^{(-0.8)} \approx 0.449$	✗	Slower
$0.80 > 1/3$	$-2 + 2.4 = +0.4$	$e^{(+0.4)} \approx 1.492$	✗	GROWING

Table 1: Numerical verification of $\varepsilon_0 = 1/3$.

6. Falsifiability and Connection to Accepted Science

Three independent falsifiability conditions:

F1: No retrocausal enrichment observed in quantum systems below $g_6 = 33$ cycles would refute T1.

F2: Absence of spiral (vs. linear) enrichment upon circuit closure in contact-geometric simulations would refute T1.

F3: Violation of dm^3 structural stability ($\epsilon_0 = 1/3$) under operator perturbations would refute T1.

Framework	What It Provides	What GTCT Adds
Two-State Vector Formalism (Aharonov & Vaidman 2001)	Forward + backward evolving states; future measurements constrain past correlations	Explicit operator algebra $G=U \circ F \circ K \circ C$ and stability threshold $g_6 = 33$
Delayed-Choice Quantum Eraser (Kim et al. 2000, PRL 84:1)	Future measurement retroactively determines past path/wave behaviour	The Fold operator F is the mathematical mechanism: post- U boundary conditions constrain prior C/K selections
Entanglement Swapping (Ma et al. 2012, Nature Physics)	Photons that never coexisted show correlations: non-local time at quantum scale	The spiral return $x_0 \rightarrow x_{64} \rightarrow x_0'$ formalises this: past is enriched, not rewritten
Negative Time Experiments (Angulo et al. 2024, arXiv:2409.03680)	Photons appear to exit medium before entering: negative group delay	Empirical signature of the Fold operator F at the quantum level
Contact Hamiltonian Mechanics (Herczeg & Waldron 2017)	Generally covariant QM on phase-spacetime using contact structures	dm^3 is the dissipative realisation with explicit limit-cycle attractors and stability radius

Table 2: GTCT confirms and extends each framework listed above.

7. Exercises / Exercícios

Six graded exercises. A student who completes all six has verified the theorem from five independent angles.

E1 [Introductory]: Verify numerically that $\log_2(3!) \cdot 4 = 10.3398\dots$ and therefore $\lceil 10.3398 \rceil = 11$. Then verify $3 \times 11 = 33$. Explain in your own words why multiplying by 3 rather than 1 is necessary for robustness.

E2 [Calculus]: Prove the Gronwall inequality in differential form using the integrating factor $\mu(t) = \exp(-\int_0^t k(s)ds)$. Apply it to the dm^3 transverse deviation $\xi(t)$ with $k = |\mu_{\max}| = 2$ and show $\epsilon_0 = 1/3$ is exact.

E3 [Differential Equations]: Solve the dm^3 normal form equations numerically (Euler method, $\Delta t = 0.01$) for $(\rho_0, \theta_0, z_0) = (1.2, 0, 0)$ with $\mu_{\max} = -2$, $\beta = 1$, $\omega = 1$. Integrate for 10 periods. Plot $\rho(t)$ and show convergence to $\Gamma = \{\rho=1\}$. Compute $\lambda_{\perp} = e^{(-4\pi)}$ and compare to the numerical rate.

E4 [Contact Geometry]: Show that the dm^3 contact form $\alpha = dz - \lambda$ is non-degenerate: $\alpha \wedge (d\alpha)^n \neq 0$ everywhere. Explain why symplectic geometry ($d\lambda$ only, no z variable) cannot support attractors on compact manifolds but contact geometry can.

E5 [Quantum Foundations]: Read Kim et al. (2000) PRL 84:1. Identify which of C, K, F, U corresponds to each stage of the delayed-choice quantum eraser experiment. Which operator is the beam-splitter choice? Which is the Fold that makes the future measurement retroactively constrain the past path? Write a 400-word analysis.

E6 [Research Level]: The GTCT predicts retrocausal enrichment only above $g_6 = 33$ cycles. Design an experimental protocol (quantum optics or NMR) to test this threshold. Specify: (1) system and rationale; (2) operationalisation of 33 cycles; (3) observable distinguishing spiral from linear return; (4) result that would falsify T1. Submit as a 1-page research proposal.

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C → K → F → U → T → ∞ → source

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Dedicated to Vic, Alice, and Sarah — once tiny, always strong.

Revised Sections for Referee Submission

Section 2 (Taxonomy Definition) · Research Directions

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These two sections replace the original Section 2 ("Derivation of the Stability Threshold $g = 33$ ") and the implicit retrocausality claims in Step 4 of the proof. All other sections of GTCT-2026-001 remain unchanged. Cross-references are updated below.

2. The g-Series: A Regime Taxonomy

The dm^3 -GTCT framework uses a taxonomy of behavioral scales called the **g-series**. This section defines the taxonomy precisely. The g-series makes no mathematical claims about the index values; the indices are labels, chosen by the criteria stated below, not derived constants.

Definition 2.1 (g-Series Taxonomy).

The g-series is a finite ordered set of regime labels

$$g^0 < g^2 < g^6 < g^{33} < g^{64}$$

Each label g^k names a qualitative behavioral regime of a dm^3 system. The superscript k is an ordinal index, not an exponent. The series is not a sequence of numbers; it is a classification.

Definition 2.2 (Index Criteria).

The five indices are assigned by the following criteria, each corresponding to a qualitatively distinct stability property of the operator chain $G = U \blacksquare F \blacksquare K \blacksquare C$:

Label	Index	Regime Name	Criterion
$g^{\blacksquare}$	0	Seed	Single operator application; no closure condition.
g^2	2	Compositional	Two-operator closure; minimal self-reference (F active).
$g^{\blacksquare}$	6	Cycle-scale	Full G-cycle closure; limit cycle Gamma entered.
g^{33}	33	Soft equilibrium	Heuristic stability index: $\text{ceil}(\log_2(3!) * 4) * 3 = 33$. See Remark 2.3.
$g^{\blacksquare\blacksquare}$	64	Phase threshold	Complete possibility-space coverage; $2^6 = 64$ operator-state combinations.

Remark 2.3 (Status of the index 33).

The index 33 arises from the following heuristic computation: there are three invariants that must close simultaneously (orthogonality, nilpotency of deviations, spectral collapse), giving $3! = 6$ orderings; $\log_2(6) \cdot 4 \approx 10.34$, so $\text{ceil}(\log_2(3!)) \cdot 4 = 11$; multiplying by 3 for a triple-confirmation robustness criterion gives $3 \times 11 = 33$. *This computation is heuristic. The "multiply by 3" step reflects a design choice — that three independent closures are required for robust stability — which is not derived from the contact-geometric structure of Sections 5–6. The index 33 is therefore a motivated label, not a theorem. Its mathematical status is that of a conjecture: we conjecture that dm^3 systems require at least 33 operator cycles for the three invariants to achieve simultaneous robust closure, and we identify this as an open verification target for the AXLE Lean 4 repository.*

Remark 2.4 (Crystal notation).

The notation $g^6:33$ and $g^{33}:33$ used elsewhere in the Principia Orthogona series denotes association, not equality. " $g^6:33$ " means: the cycle-scale regime g^6 carries the structural index 33. It is a descriptor of the same kind as " D_6 symmetry: 6-fold" or " E_8 lattice: rank 8". No algebraic identity $g^6 = 33$ is claimed.

Cross-reference update. All references in Sections 3–6 to "the derivation of $g = 33$ " should be read as references to Definition 2.2 and Remark 2.3 above. The stability radius $\epsilon_0 = 1/3$ (Section 5) is independent of the g -series taxonomy and remains a theorem.

8. Research Directions

This section replaces the retrocausality claims in Step 4 of the original proof. The contact-geometric results of Sections 5–6 are classical dissipative dynamics; they do not imply retrocausality. The connections described below are structural analogies and open research problems, not theorems.

8.1 Operator Grammar and Time-Symmetric Quantum Mechanics

The two-state vector formalism (TSVF) of Aharonov and Vaidman [4] assigns both a forward-evolving and a backward-evolving state to each quantum system. The operator chain $G = U \blacksquare F \blacksquare K \blacksquare C$ has a structural resemblance to this bidirectional picture: the Fold operator F introduces self-reference at mid-circuit, and the Unfold operator U projects outward. Whether this resemblance can be made precise — by constructing a map from the dm^3 contact system to the TSVF Hilbert-space formalism — is an open problem. No retrocausal claim is made in this paper.

Open problem 8.1. Construct, if possible, a functor from the dm^3 operator category to the category of pre- and post-selected quantum states in the TSVF, preserving the stability radius $\epsilon_0 = 1/3$.

8.2 Delayed-Choice Experiments and the Fold Operator

In the delayed-choice quantum eraser (Kim et al. [5]), a future measurement choice determines which past path information is available. The Fold operator F — which introduces the first self-reference in the chain — is structurally analogous to the beam-splitter choice in that experiment: it is the point at which the system curves back onto its own prior state. This analogy motivates the following research direction, but does not constitute a proof of any physical connection.

Open problem 8.2. Identify an experimental observable in a quantum optics or NMR system that would distinguish "spiral enrichment" ($x' \neq x$ after circuit closure) from linear return ($x' = x$), and determine whether the dm^3 stability threshold g^{33} has a measurable counterpart in that system.

8.3 Entanglement Swapping and the Spiral Return

Ma et al. [6] demonstrated entanglement swapping between photons that never coexisted, suggesting non-trivial temporal structure at the quantum scale. The spiral return $x \rightarrow x'' \rightarrow x'$ with $x' \neq x$ in the dm^3 circuit is formally analogous: the source state is enriched by one completed circuit without returning to its original configuration. Whether this formal analogy corresponds to a physical mechanism is unknown.

Open problem 8.3. Determine whether the contact-geometric spiral return, formalized via the Floquet multiplier $\lambda_{\text{transverse}} = e^{-4\pi i}$, admits a quantum information interpretation consistent with the entanglement swapping results of Ma et al.

8.4 Negative Group Delay and the Weave Scale

Angulo et al. [7] report experimental evidence of negative group delay — photons appearing to exit a medium before entering. This is a well-understood phenomenon in classical wave optics (anomalous dispersion) and does not require retrocausality. The dm^3 framework does not explain or predict negative group delay. The reference is included as context for the broader research landscape, not as evidence for the operator grammar.

8.5 Formal Verification in AXLE (Lean 4)

The AXLE repository (github.com/TOTOGT/AXLE) contains Lean 4 proof obligations for the dm^3 framework. The highest-priority open items relevant to this paper are:

- (a) Formal verification of the Gronwall bound and $\varepsilon_0 = 1/3$ (Section 5) — currently the most tractable target.
- (b) Formalization of the g-series taxonomy (Definition 2.2) as a Lean 4 inductive type, with the index-33 conjecture stated as a sorry to be discharged by future work.
- (c) The Poincare-to-Collatz correspondence — the hardest open item, currently marked as a sorry in AXLE — is not required for the results of this paper.

Note on References

References [4]–[8] (Aharonov-Vaidman, Kim et al., Ma et al., Angulo et al., Herczeg-Waldron) are retained in the bibliography as motivation for the research directions in Section 8. None of these references is cited as support for a theorem in this paper. The causal and retrocausal claims in Step 4 of the original proof are withdrawn and replaced by Open Problems 8.1–8.3 above.